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AEROSOL GENERATING DEVICE AND HEATING ASSEMBLY

Abstract

The present disclosure relates to an aerosol generating device and a heating assembly. The heating assembly includes a heating portion configured to generate infrared light waves in a power-on state and a sleeve for the infrared light waves to pass through. The heating portion includes a heating substrate, an antioxidant layer arranged on the outer surface of the heating substrate and configured to prevent the heating substrate from being oxidized, and an infrared radiation layer arranged on the side of the antioxidant layer away from the heating substrate. An accommodating cavity configured to accommodate the heating portion and non-sealed is formed in the sleeve. In the heating assembly, with the antioxidant layer, the sleeve's accommodating cavity doesn't require sealing, vacuuming, or inert gas filling. This simplifies the assembly process and cuts manufacturing costs.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/114120, filed on Aug. 21, 2023, which claims priority to Chinese Patent Application No. 202211458267.6, filed on Nov. 17, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD

[0002] The present disclosure relates to the field of heat-not-burn atomization, and in particular, to an aerosol generating device and a heating assembly.

BACKGROUND

[0003] In the field of HNB (heat-not-burn) atomization, heating methods such as heating through a central heating element heating or a circumferential heating element are generally adopted. The usual approach is that the heating element generates heat, and then the heat is directly transferred to an aerosol generating substrate through thermal conduction. The substrate is usually atomized within 350° C. The disadvantage of this heating method is that since the heating element conducts heat directly or indirectly through a solid material to the aerosol generating substrate, it requires that the working temperature of the heating element should not be too high, and otherwise it will cause the aerosol generating substrate to be overheated or the solid material to produce a peculiar smell to influence the puffing taste of the electronic cigarette.

[0004] In the related arts, since the heating assembly that generates thermal radiation for heating is adopted and the working temperature of the heating element may reach around 400° C., when the heating element works at high temperatures, its substrate material will have the problem of oxidation, resulting in significant changes in the resistance value of the heating element and affecting its heating stability. To solve the problem of oxidation, the sealed mounting space is generally vacuumed or filled with inert gas. The process is not only complex but also has a high manufacturing cost.

SUMMARY

[0005] In an embodiment, the present invention provides a heating assembly, comprising: a heating portion configured to generate infrared light waves in a power-on state; and a sleeve for the infrared light waves to pass through, wherein the heating portion comprises a heating substrate, wherein an antioxidant layer is arranged on an outer surface of the heating substrate and configured to prevent the heating substrate from being oxidized, wherein an infrared radiation layer is arranged on a side of the antioxidant layer away from the heating substrate, wherein an accommodating cavity configured to accommodate the heating portion and is formed in the sleeve, the accommodating cavity being non-sealed, and wherein the heating portion is at least partially spaced apart from a tube wall of the sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or

combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0007] FIG. 1 is a schematic exploded structural diagram of an aerosol generating device according to a first embodiment of the present disclosure.

[0008] FIG. 2 is a schematic structural diagram of a heating assembly of the aerosol generating device in FIG. 1.

[0009] FIG. 3 is a first longitudinal cross-sectional diagram of the heating assembly in FIG. 2.

[0010] FIG. 4 is a second longitudinal cross-sectional diagram of the heating assembly in FIG. 2.

[0011] FIG. 5 is a third longitudinal cross-sectional diagram of the heating assembly in FIG. 2.

[0012] FIG. 6 is a schematic exploded structural diagram of the heating assembly in FIG. 2.

[0013] FIG. 7 is a schematic bottom structural diagram of the heating assembly in FIG. 2.

[0014] FIG. 8 is a transverse cross-sectional diagram of a heating element of the heating assembly in FIG. 6.

[0015] FIG. 9 is a transverse cross-sectional diagram of a heating element of an aerosol generating device according to a second embodiment of the present disclosure.

[0016] FIG. 10 is a transverse cross-sectional diagram of a heating element of an aerosol generating device according to a third embodiment of the present disclosure.

DETAILED DESCRIPTION

[0017] In an embodiment, the present invention provides an improved heating assembly and further provide an improved aerosol generating device.

[0018] In an embodiment, the present invention provides a heating assembly, including a heating portion configured to generate infrared light waves in a power-on state and a sleeve for the infrared light waves to pass through.

[0019] The heating portion includes a heating substrate, an antioxidant layer arranged on the outer surface of the heating substrate and configured to prevent the heating substrate from being oxidized, and an infrared radiation layer arranged on the side of the antioxidant layer away from the heating substrate; and [0020] an accommodating cavity configured to accommodate the heating portion and non-sealed is formed in the sleeve, and the heating portion is at least partially spaced apart from the tube wall of the sleeve.

[0021] In some embodiments, the antioxidant layer includes an oxide film, and the oxide film is formed on the outer surface of the heating substrate.

[0022] In some embodiments, the thickness of the antioxidant layer is 1 μm -150 μm .

[0023] In some embodiments, an air gap is provided between the inner wall of the accommodating cavity and the heating portion.

[0024] In some embodiments, the sleeve includes a hollow tubular body; [0025] the accommodating cavity is formed in the tubular body; and [0026] one end of the tubular body is provided with an opening.

[0027] In some embodiments, the heating portion is connected to two conductive portions; and [0028] the two conductive portions penetrate through the opening.

[0029] In some embodiments, the sleeve includes a pointed structure, and the pointed structure is arranged at the end of the tubular body away from the opening.

[0030] In some embodiments, the number of the conductive portions is two, and the two conductive portions are spaced apart; and [0031] the heating assembly further includes an insulating member at least partially arranged in the sleeve and configured to insulate the two conductive portions.

[0032] In some embodiments, the heating assembly further includes a supporting base configured to support the heating portion and the sleeve; and the sleeve is at least partially inserted into the supporting base.

[0033] In some embodiments, a conductive member connected to the conductive portions is

provided in the supporting base.

[0034] In some embodiments, the supporting base includes a holder configured to support the sleeve, and a seal member; and [0035] the seal member is sleeved over a partial section of the sleeve and seals a gap between the inner wall of the holder and the outer wall of the sleeve.

[0036] In some embodiments, the seal member is a hollow structure with two ends running through, and a channel for the sleeve to pass through is formed on the inner side.

[0037] In some embodiments, the seal member includes a sleeve body with two ends running through and configured to be partially sleeved over the sleeve, and a first seal portion protruding from the outer side wall of the sleeve body; and [0038] the first seal portion is clamped and fixed with the holder.

[0039] In some embodiments, the supporting base includes a shell sleeved over the outer periphery of the holder and provided with a sleeve opening for cooperating with the holder; and [0040] the holder includes the bottom wall, and a gap is provided between the sleeve opening and the bottom wall.

[0041] In some embodiments, the shell is detachably sleeved over the holder; and [0042] the shell is provided with a through hole for the heating structure to partially penetrate through.

[0043] In some embodiments, the seal member further includes a sleeve body with two ends running through and configured to be partially sleeved over the sleeve, and a second seal portion protruding from the outer side wall of the sleeve body; and the second seal portion is located between the holder and the shell in an assembled state of the shell and the holder, and is configured to seal a gap formed between the holder and the end surface of the through hole.

[0044] In some embodiments, the sleeve is made of infrared transparent glass, transparent ceramics, or diamond.

[0045] In some embodiments, the heating element is spaced apart from the tube wall of the sleeve on the whole.

[0046] In some embodiments, the heating element is arranged without being in direct contact with the sleeve.

[0047] In some embodiments, the infrared radiation layer includes an infrared layer and/or a composite infrared layer, and the composite infrared layer is formed by compounding an infrared layer forming substrate with a bond configured to bond with the antioxidant layer.

[0048] In some embodiments, the heating substrate includes a metal substrate; and the metal substrate includes a nickel-chromium alloy substrate [0049] or an iron-chromium-aluminum alloy substrate.

[0050] The present disclosure further provides an aerosol generating device, including the heating assembly described above.

[0051] The aerosol generating device and the heating assembly in the present disclosure have the following beneficial effects: In the heating assembly, by arranging the antioxidant layer on the outer surface of the heating substrate of the heating portion, the heating substrate can be prevented from being oxidized, so that the accommodating cavity for accommodating the heating portion in the sleeve can be non-sealed, that is, the sleeve does not need to be sealed, vacuumed, or filled with inert gas, and the space in the sleeve can be communicated with the atmosphere outside the instrument, thereby simplifying the assembling process of the heating assembly and reducing the manufacturing cost.

[0052] In addition, by arranging the infrared radiation layer on the outer surface of the heating substrate, when the heating substrate generates heat in a power-on state, the heat can excite the infrared radiation layer to radiate infrared light waves, and the infrared light waves can pass through the sleeve to reach and heat the aerosol generating substrate. In a case that the highest working temperature of the heating element reaches 1000° C. or above (the working temperature of traditional HNB heating elements generally does not exceed 400° C.), it will not cause the aerosol generating substrate to be overheated, thereby greatly improving the puffing taste. Moreover, in a

high-temperature working state, the preheating time is greatly reduced, thereby greatly improving the consumer experience.

[0053] To provide a clearer understanding of the technical features, objectives, and effects of the present disclosure, the specific embodiments of the present disclosure will be described below with reference to the drawings.

[0054] FIG. 1 illustrates an aerosol generating device according to a first embodiment of the present disclosure. The aerosol generating device **100** may adopt a low-temperature heat-not-burn method to heat an aerosol generating substrate, and has the advantages of good atomization stability and excellent puffing taste. In some embodiments, the aerosol generating substrate may be arranged in the aerosol generating device **100** in a pluggable manner. The aerosol generating substrate may be cylindrical. Specifically, the aerosol generating substrate may be a filamentous or sheet-like solid material made from leaves and/or stems of plants, and aroma components may be further added to the solid material.

[0055] Referring to FIG. 1 and FIG. 2, further, in this embodiment, the aerosol generating device **100** includes a heating assembly **10** and a power supply assembly **20**. The power supply assembly **20** includes a power supply case **21**. The heating assembly is accommodated in the power supply case **21** and may be partially inserted into the aerosol generating substrate. Specifically, it may be partially inserted into a substrate section of the aerosol generating substrate, and generates infrared radiation to heat the substrate section of the aerosol generating substrate in a power-on state, causing it to be atomized to generate an aerosol. The heating assembly **10** has the advantages of easiness in assembling, simple structure, high atomization efficiency, strong stability, and long service life. The power supply assembly **20** is mechanically and/or electrically connected to the heating assembly **10** and is configured to supply power to the heating assembly **10**.

[0056] In this embodiment, the heating assembly **10** includes a heating structure **11** and a supporting base **12**. The heating structure **11** is mounted on the supporting base **12**. In this embodiment, the heating structure **11** and the supporting base **12** are detachably mounted to facilitate the replacement and maintenance of the heating structure **11**. The supporting base **12** may be mechanically and electrically connected to the heating structure **11**. It can not only support the heating structure **11**, but also be electrically connected to the heating structure **11** when the heating structure **11** is mounted on it, thereby electrically connecting the heating structure **11** to the power supply assembly **20**. As can be understood, in some other embodiments, the supporting base **12** may only play a supporting role.

[0057] Referring to FIG. 3 to FIG. 5, in this embodiment, the heating structure **11** includes a sleeve **111** and a heating element **112**. The sleeve **111** is at least partially inserted into the supporting base **12**, covers at least part of the heating element **112**, and may allow light waves to pass through to reach the aerosol generating substrate. Specifically, in this embodiment, the sleeve **111** may allow infrared light waves to pass through, thereby facilitating the heating element **112** to radiate heat to heat the aerosol generating substrate. Specifically, in this embodiment, the heating element **112** is spaced apart from the tube wall of the sleeve **111** on the whole, and a gap is provided between the inner wall of the sleeve **111** and the heating element **112**. In a power-on state, the heating element quickly heats to increase the temperature up to 1000-1300° C. within 1-3 s, the surface temperature of the sleeve **111** can be controlled below 350° C., and the overall atomization temperature of the aerosol generating substrate is controlled at 300-350° C., thereby achieving the precise atomization of the aerosol generating substrate mainly in a wave band of 2-5 μm . In the present disclosure, the highest working temperature of the heating element ranges from 500° C. to 1300° C., which is much higher than the highest working temperature of the heating element in the existing technology.

[0058] In this embodiment, the sleeve **111** may be a quartz glass tube. Of course, as can be understood, in some other embodiments, the sleeve **111** is not limited to a quartz tube and may be any other window material that allows light waves to pass through, such as infrared transparent

glass, transparent ceramics, or diamond.

[0059] Referring to FIG. 6 to FIG. 8, in this embodiment, the sleeve **111** is a hollow tubular body. Specifically, the sleeve **111** includes a tubular body **1111** with a circular cross section, and a pointed structure **1112** arranged at one end of the tubular body **1111**. Of course, as can be understood, in some other embodiments, the cross section of the tubular body **111** is not limited to be circular. The tubular body **1111** is a hollow structure with an opening **1110** in one end. The pointed structure **1112** is arranged at the end of the tubular body **1111** away from the opening **1110**. By arranging the pointed structure **1112**, it is convenient for at least part of the heating structure **111** to be plugged into the aerosol generating substrate. In this embodiment, an accommodating cavity **1113** is formed on the inner side of the sleeve **111**. The accommodating cavity **1113** is a cylindrical cavity and may be non-sealed. The air in the accommodating cavity **1113** may be communicated with the atmosphere outside the instrument. When the heating element **112** is mounted in it, the accommodating cavity **1113** does not need to be vacuumed or filled with inert gas. In this embodiment, the sleeve **111** further includes a positioning portion **1114**. The positioning portion **1114** is arranged at the opening **1110** of the tubular body **111** and may extend radially outwards along the tubular body **111** to form a positioning flange for the mounting and positioning of the sleeve **111** and the supporting base **12**. In this embodiment, the positioning portion **1114** may be integrally formed with the tubular body **111**. Of course, as can be understood, in some other embodiments, the positioning portion **1114** may be detachably assembled with the sleeve **111** by sleeving, screwing, or clamping, for example. In this embodiment, an air gap is provided between the inner wall of the sleeve **111** and the heating element **112**. The air gap may be filled with air. By providing the air gap, there can be no direct contact between the sleeve **111** and the heating element **112**.

[0060] In this embodiment, the number of the heating element **112** may be one, which may be longitudinally arranged and wound to form a heating portion **1120** which is spiral on the whole. Specifically, the heating element **112** may be cylindrical on the whole and may be wound to form a single-spiral structure, double-spiral structure, M-shaped structure, N-shaped structure, or structure with any other shape. Of course, as can be understood, in some other embodiments, the number of the heating element **112** is not limited to one, and may be two or more. The shape of the heating element **112** is not limited to a cylindrical shape. In some embodiments, the shape of the heating element **112** may be a sheet shape.

[0061] In this embodiment, the heating portion **1120** may be placed in the sleeve **111**, spaced apart from the inner wall of the sleeve **111**, and configured to generate infrared radiation, that is, generate infrared light waves, in a power-on state. The infrared light waves may pass through the sleeve **111** to reach the aerosol generating substrate. In this embodiment, the heating portion **1120** may be in a longitudinally elongated spiral shape. Of course, as can be understood, in some other embodiments, the heating portion **1120** is not limited to a spiral shape.

[0062] In this embodiment, one end of the heating portion **1120** is provided with a conductive portion **1121**. The conductive portion **1121** is connected to the heating portion **1120**, and may be led out from the opening **1110** of the sleeve **111**, penetrate through a base, and be electrically connected to the power supply assembly **20**. In this embodiment, the conductive portion **1121** may be fixed to the heating portion **1120** through welding to form an integrated structure. Of course, as can be understood, in some other embodiments, the heating portion **1120** may be integrally formed with the conductive portion **1121**. In this embodiment, the number of the conductive portions **1121** may be two. The two conductive portions **1121** may be spaced apart, be respectively connected to two ends of the heating portion **1120**, extend towards the same end, and penetrate through the sleeve **111** from the opening **1110** in one end of the sleeve **111**. In this embodiment, the conductive portion **1121** may be a lead wire, which may be welded to the heating portion **1120**. Of course, as can be understood, in some other embodiments, the conductive portion **1121** is not limited to a lead wire and may be any other conductive structure. By arranging the conductive portion **1121** at one end of

the heating portion **1120** and leading it out from the sleeve **111**, the entire heating structure **11** can be conveniently assembled, thereby simplifying the assembling process. During assembling, the heating structure **11** may be mounted on the supporting base **12**, and then enabled to be in contact with a conductive member **124** located in the supporting base **12**.

[0063] In this embodiment, the heating element **112** forming the heating portion **1120** includes a heating substrate **1122** and an infrared radiation layer **1124**. The heating substrate **1122** may generate heat in a power-on state. The infrared radiation layer **1124** is arranged on the outer surface of the heating substrate **1122**, and is heated through the heating substrate **1122** to excite and radiate infrared light waves. In this embodiment, the heating substrate **1122** and the infrared radiation layer **1124** are concentrically distributed on the cross section of the heating portion **1120**.

[0064] In this embodiment, the heating substrate **1122** may be cylindrical on the whole. Specifically, the heating substrate **1122** may be a heating wire. Of course, as can be understood, in some other embodiments, the heating substrate **1122** may not be limited to be cylindrical, but may be in a sheet shape, that is, the heating substrate **1122** may be a heating sheet. The heating substrate **1122** includes a metal substrate with high-temperature oxidation resistance. The metal substrate may be a metal wire. Specifically, the heating substrate **1122** may be a nickel-chromium alloy substrate (such as nickel-chromium alloy wire), iron-chromium-aluminum alloy substrate (such as iron-chromium-aluminum alloy wire) or the like made of a metal material with good high-temperature oxidation resistance, high stability, and good deformation resistance. In this embodiment, the radial size of the heating substrate **1122** may be 0.15 mm-0.8 mm.

[0065] In this embodiment, the heating element **112** further includes an antioxidant layer **1123**, and the antioxidant layer **1123** is formed between the heating substrate **1122** and the infrared radiation layer **1124**. Specifically, the antioxidant layer **1123** may be an oxide film, the heating substrate **1122** undergoes high-temperature heat treatment to form a dense oxide film on its own surface, and the oxide film forms the antioxidant layer **1123**. Of course, as can be understood, in some other embodiments, the antioxidant layer **1123** is not limited to include an oxide film formed by itself. In some other embodiments, it may be an antioxidant coating layer coated on the outer surface of the heating substrate **1122**. By forming the antioxidant layer **1123**, it can ensure that the heating substrate **1122** is not or is rarely oxidized when heated in the air environment, thereby improving the stability of the heating substrate **1122**. Therefore, there is no need to vacuum, fill inert gas or reducing gas into the accommodating cavity **1113**, nor to block the opening **1110**, thereby simplifying the assembling process of the entire heating structure **11** and reducing the manufacturing cost. In this embodiment, the thickness of the antioxidant layer **1123** may be selectively 1 μm -150 μm . When the thickness of the antioxidant layer **1123** is less than 1 μm , the heating substrate **1122** is easily oxidized. When the thickness of the antioxidant layer **1123** is greater than 150 μm , it will influence the heat conduction between the heating substrate **1122** and the infrared radiation layer **1124**.

[0066] In this embodiment, the infrared radiation layer **1124** may be an infrared layer. The infrared layer may be an infrared layer forming substrate formed on the side of the antioxidant layer **1123** away from the heating substrate **1122** under high-temperature heat treatment. In this embodiment, the infrared layer forming substrate may be a silicon carbide, spinel, or composite substrate thereof. Of course, as can be understood, in some other embodiments, the infrared radiation layer **1124** is not limited to an infrared layer. In some other embodiments, the infrared radiation layer **1124** may be a composite infrared layer. In this embodiment, the infrared layer may be formed on the side of the antioxidant layer **1123** away from the heating substrate **1122** through dip coating, spray coating, brush coating, and other methods. The thickness of the infrared radiation layer **1124** may be 10 μm -300 μm . When the thickness of the infrared radiation layer **1124** is between 10 μm -300 μm , the thermal radiation effect is better, and the atomization efficiency and taste of the aerosol generating substrate are better. Of course, as can be understood, in some other embodiments, the thickness of the infrared radiation layer **1124** is not limited to 10 μm -300 μm .

[0067] In this embodiment, the heating assembly **11** further includes an insulating member **113**, the insulating member **113** is cylindrical, and its radial size may be smaller than the radial size of the accommodating cavity **1113**. The insulating member **113** may fully or partially penetrate into the accommodating cavity **1113** through the opening **1110** of the sleeve **111**, thereby separating the two conductive portions **1121**, that is, insulating the two conductive portions **1121**. In this embodiment, the insulating member **113** is provided with through holes **1131**, the number of the through holes **1131** is two, the two through holes **1131** correspond to the two conductive portions **1121** one to one, and the through holes **1131** may extend along the axial direction of the insulating member **113** and be configured to allow the conductive portions **1121** to penetrate through for electrical connection with the supporting base **12**. In some embodiments, the insulating member **113** may not be limited to be cylindrical. In some embodiments, the insulating member **113** may be an insulating partition, and the through holes **1131** may be omitted. In some embodiments, the insulating member **113** may be a ceramic body, quartz tube, or any other insulating structure.

[0068] Further referring to FIG. 3 to FIG. 7, in this embodiment, the supporting base **12** may support the sleeve **111** and the heating portion **1120**, and include a holder **121**, a shell **122**, and a seal member **123**. The holder **121** is configured to support the heating structure **11**. The shell **122** may be sleeved over the outer periphery of the holder **121**. The seal member **123** may be mounted on the holder **121** and configured to hermetically connect the heating structure **11** with the holder **121** and the shell **122**.

[0069] In this embodiment, the holder **121** includes a first holder body **121a** and a second holder body **121b** that are openable and closable. By arranging the first holder body **121a** and the second holder body **121b** that are openable and closable, the mounting and dismounting of the heating structure **11** can be facilitated. In some embodiments, splicing the first holder body **121a** and the second holder body **121b** can form a rectangular solid structure. Of course, as can be understood, in some other embodiments, it is not limited to form a rectangular solid shape by splicing the first holder body **121a** and the second holder body **121b**. In some other embodiments, splicing the first holder body **121a** and the second holder body **121b** may form a cylindrical or other shape.

[0070] In this embodiment, one end of each of the first holder body **121a** and the second holder body **121b** is provided with an end plate **1210**, separating plates **1212** are correspondingly arranged in both the first holder body **121a** and the second holder body **121b**, the separating plates **1212** divide the holder **121** into upper and lower spaces, a clamping groove **1211** for cooperating with the seal member **123** is formed in the space close to the end plate **1210**, each separating plate **1212** is provided with a semi-cylindrical first avoidance hole **1216**, and the two separating plates **1212** of the first holder body **121a** and the second holder body **121b** are arranged opposite to each other, and the first avoidance holes **1216** are spliced to form a first through hole for the heating structure **11** to pass through.

[0071] In this embodiment, the holder **121** further includes the bottom wall **1213**, and the bottom wall **1213** is arranged on the first holder body **121a**. Of course, as can be understood, in some other embodiments, the bottom wall **1213** is not limited to be arranged on the first holder body **121a**, but may also be arranged on the second holder body **121b**.

[0072] In this embodiment, a separating member **1215** is arranged in the supporting base **12**. Specifically, the separating member **1215** protrudes from the bottom wall **1213** and is integrally formed with the bottom wall **1213**. It may be a rib plate and be configured to separate the two adjacent conductive portions **1121** and insulate the two conductive portions **1121**.

[0073] In this embodiment, the first holder body **121a** and the second holder body **121b** are provided with limiting plate **1216** for limiting the heating structure **11**, the limiting plates **1216** are arranged below the separating plates **1212** and are spaced apart from the separating plates **1212**, and each limiting plate **1216** is provided with a semi-cylindrical second avoidance hole **1217**. When the first holder body **121a** and the second holder body **121b** are spliced, the two second avoidance holes **1217** in the two limiting plates **1216** are spliced to form a second through hole for

the heating structure **11** to pass through. The radial size of the second through hole is smaller than the radial size of the positioning portion **1114** at one end of the sleeve **111**, thereby cooperating with the positioning portion **1114** to position the heating structure **11**.

[0074] In this embodiment, the shell **122** may be sleeved over the outer periphery of the holder **121** after the heating structure **11** is assembled with the holder **121**, thereby playing a role of fixing the first holder body **121a** and the second holder body **121b**, so that the heating structure **11** and the supporting base **12** form an integrated structure. In this embodiment, the shape and size of the shell **122** may be adapted to the holder **121**. In this embodiment, the shell **122** is roughly in a rectangular solid shape and is a hollow structure with a sleeve opening **1221** in one end. A gap **1220** is provided between the sleeve opening **1221** and the bottom wall **1213**, thereby preventing the residual aerosol in the power supply case **21** from condensing and forming condensate that affects the normal operation of the heating structure **11**.

[0075] In this embodiment, the shell **122** may be detachably connected with the holder **121**. Specifically, in this embodiment, [0076] the shell **122** and the holder **121** are provided with a connecting structure **125**, and the shell **122** and the holder **121** are detachably connected through the connecting structure **125**. In this embodiment, the connecting structure **125** includes clamping holes **1222** and clamping buckles **1214**. The clamping buckles **1214** protrude from the outer side wall of the holder **121**. Specifically, the number of the clamping buckles **1214** is two. The two clamping buckles **1214** are correspondingly arranged on the outer side walls of the first holder body **121a** and the second holder body **121b**. The clamping holes **1222** are provided in the side wall of the shell **122** and the number is two. The two clamping holes **1222** correspond to the two clamping buckles **1214** one to one. When the shell **122** is assembled with the holder **121**, the clamping buckles **1214** can be clamped into the clamping holes **1222**, thereby connecting and fixing the shell **122** and the holder **121**.

[0077] In this embodiment, the side of the shell **122** opposite to the sleeve opening **1221** is provided with a blocking wall **1223**, and the shell **122** is provided with a through hole **1224**. Specifically, the through hole **1224** is provided in the blocking wall **1223** and can allow the heating structure **11** to partially penetrate through.

[0078] In this embodiment, the seal member **123** is detachably arranged between the first holder body **123a** and the second holder body **123b**, and the seal member **123** is detachably sleeved over the heating structure **11**. Specifically, it may be sleeved over the outer periphery of a partial section of the sleeve **111**, and configured to hermetically connect the heating structure **11** with the first holder body **121a** and the second holder body **121b**. In this embodiment, the seal member **123** may be a silica gel member, which serves to prevent vibration and prevent damage when the sleeve **111** and the holder **121** are assembled. Of course, as can be understood, in some other embodiments, the seal member **123** is not limited to a silica gel member.

[0079] In this embodiment, the seal member **123** is a hollow structure with two ends running through, a channel **1230** may be formed on the inner side, and the channel **1230** may allow the sleeve **11** to pass through. In this embodiment, the seal member **123** includes a sleeve body **1231**, a first seal portion **1232**, and a second seal portion **1233**. The sleeve body **1231** is cylindrical, is a hollow structure with two ends running through, and is configured to be sleeved over part of the heating structure **11**. The first seal portion **1232** and the second seal portion **1233** protrude from the outer side wall of the sleeve body **1231**, and are spaced apart along the axial direction of the sleeve body **1231**. The first seal portion **1232** may be arranged along the circumferential direction of the sleeve body **1232**, and may be roughly in a circular ring shape. The first seal portion **1232** is respectively clamped and fixed with the first holder body **121a** and the second holder body **121b**. Specifically, the first seal portion **1232** may be respectively clamped into the clamping grooves **1211** in the first holder body **121a** and the second holder body **121b**. The second seal portion **1233** protrudes from the outer side wall of the sleeve body **1231** and is roughly in a circular ring shape, and its radial size is larger than the radial size of the first seal portion **1232**. The second seal portion

1233 may be placed on the side where the end wall **1210** of the first holder body **121a** and the second holder body **121b** is opposite to the clamping groove **1211**. In an assembled state of the shell **12** and the holder **121**, the second seal portion **1233** is located between the shell **122** and the holder **121**. Specifically, it is located between the blocking wall **1223** and the end wall **1210**, and is configured to seal a gap formed between the holder **121** and the end surface of the through hole **1224**. In this embodiment, the sleeve body **1231**, the first seal portion **1232**, and the second seal portion **1233** are an integrally formed structure, forming multiple sealing structures. That is, by arranging the seal member **123**, the sealing among the shell **122**, the heating structure **11**, and the holder **121** can be achieved, thereby simplifying the sealing process, reducing the manufacturing cost, and preventing condensate from flowing into the holder **121**.

[0080] In this embodiment, multiple conductive members **124** are arranged on the supporting base **12**. Specifically, the conductive members **124** are arranged corresponding to the conductive portions **1121** one to one. Of course, as can be understood, in some other embodiments, the number of the conductive member **124** may be one. The conductive member **124** may be an electrode pillar. The multiple conductive members **124** are spaced apart on the bottom wall **1213**, and may be detachably connected to the conductive portions **1121**. Specifically, in a state that the heating structure **11** is mounted on the supporting base **12**, the conductive portion **1121** may be wound around the conductive member **124**, thereby being electrically connected to the conductive member **124**. In this embodiment, the conductive member **124** may be electrically connected with a power supply in the power supply assembly **20** through contact, thereby electrically connecting the heating structure **11** with the power supply assembly **20** and facilitating the replacement of the heating element **122** when the heating element **112** reaches its service life. In this embodiment, the number of the conductive members **124** is two groups, one group is electrically connected to the heating structure **11**, and the other group may be connected to a temperature measurement structure **13**. Of course, as can be understood, in some other embodiments, the number of the conductive member **124** may be one group, and the heating structure **11** and the temperature measurement structure **13** may share one group of conductive members **124**.

[0081] In this embodiment, the heating assembly **10** further includes a temperature measurement structure **13**. The temperature measurement structure **13** is arranged on the heating structure **11** and may be detachably connected to the supporting base **12**. In this embodiment, the temperature measurement structure **13** may be sleeved over the outer periphery of a partial section of the sleeve **111**, and may be detachably connected to the conductive member **124** in the supporting base **12** and electrically connected to it when connected thereto. In this embodiment, the temperature measurement structure **13** is sleeved over a position on the sleeve **111** corresponding to a position of connection between the heating portion **1120** and the conductive portions **1121**, and includes a temperature measurement film **131** and lead wires **132**. The temperature measurement film **131** may be sleeved over the outer side wall of the sleeve **111**. The number of the lead wire **132** is two. The two lead wires **132** are spaced apart, one end of each lead wire is connected to the temperature measurement film **131**, the other end of each lead wire may be connected to the conductive member **132** in the supporting base **12**, and they may be wound around the corresponding conductive members **132** for electrical connection and signal transmission. In some embodiments, the lead wires **132** may be connected to the temperature measurement film **131** by welding or crimping.

[0082] When the heating assembly **10** is assembled, the temperature measurement structure **13** may be first sleeved over the outer periphery of the sleeve **111**, and then the seal member **123** is sleeved over the sleeve **111** of the heating structure **11**. The first holder body **121a** is clamped at the first seal portion **1232** of the seal member **123**. Then, the conductive portions **1121** of the heating structure **11** and the lead wires **132** of the temperature measurement structure **13** are wound around the corresponding conductive members **124**. Then, the second holder body **121b** is clamped at the second seal portion **1232**. Finally, the overall structure formed by the holder **121** and the heating

structure **11** is inserted into the sleeve opening **1221** in the shell **122**, so that the second seal portion **1233** is pressed against the blocking wall **1223** of the shell **122** and the end wall **1210** of the holder **121**, the seal member **123** and the heating structure **11** are enabled to partially penetrate through the through hole **1224**, and the clamping buckles **1214** on the outer side of the holder **121** are clamped into the sleeve opening **1221** in the shell **122**. If the heating structure **11** needs to be disassembled, the shell **122** is pushed out towards the direction of the pointed structure **1112** of the heating structure **11**, then the first holder body **121a** and the second holder body **121b** are separated from the seal member **123**, and finally the conductive portions **1121** and the lead wires **132** of the temperature measurement structure **13** are disconnected from the conductive members **124**.

[0083] FIG. **9** shows an aerosol generating device according to a second embodiment of the present disclosure, the difference between the second embodiment and the first embodiment lies in that the infrared radiation layer **1124** is a composite infrared layer. The composite infrared layer may be formed by compounding an infrared layer forming substrate with a bond configured to bond with the antioxidant layer **1123**. Specifically, the bond may be glass powder. The composite infrared layer may be a glass powder composite infrared layer. The reason for adopting glass powder is that glass powder can melt at high temperatures to bond the antioxidant layer **1123** with the infrared layer forming substrate and block the gap in the infrared layer forming substrate, thereby further improving the breakdown resistance. The glass powder composite infrared layer may be prepared by adding glass powder to the infrared layer forming substrate (such as silicon carbide or spinel), performing compounding, then coating the mixture onto the side of the antioxidant layer **1123** away from the heating substrate **1122** through dip coating, spray coating or brush coating, performing heat treatment for 30 min in a tunnel furnace, then placing the obtain material in a heating furnace, increasing the temperature to 1000-1200° C., preserving the heat for 2 h, and then cooling to room temperature with the furnace.

[0084] FIG. **10** shows an aerosol generating device according to a third embodiment of the present disclosure, the difference between the third embodiment and the first embodiment lies in that the heating element **112** further includes a bonding layer **1125** arranged between the antioxidant layer **1123** and the infrared radiation layer **1124**. The bonding layer **1125** may be configured to prevent the heating substrate **1122** from being broken down locally, thereby further improving the bonding strength between the antioxidant layer **1123** and the infrared radiation layer **1124**. In some embodiments, the bond in the bonding layer **1125** may be glass powder, that is, the bonding layer **1125** may be a glass powder layer.

[0085] In some embodiments, the bond may also be added to the infrared radiation layer **1124**, and the melting point of the glass powder selectable for the bonding layer **1125** is higher than the melting point of the glass powder in the infrared radiation layer **1124**.

[0086] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0087] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C,

regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A heating assembly, comprising: a heating portion configured to generate infrared light waves in a power-on state; and a sleeve for the infrared light waves to pass through, wherein the heating portion comprises a heating substrate, wherein an antioxidant layer is arranged on an outer surface of the heating substrate and configured to prevent the heating substrate from being oxidized, wherein an infrared radiation layer is arranged on a side of the antioxidant layer away from the heating substrate, wherein an accommodating cavity configured to accommodate the heating portion and is formed in the sleeve, the accommodating cavity being non-sealed, and wherein the heating portion is at least partially spaced apart from a tube wall of the sleeve.
2. The heating assembly of claim 1, wherein the antioxidant layer comprises an oxide film, and wherein the oxide film is formed on an outer surface of the heating substrate.
3. The heating assembly of claim 1, wherein a thickness of the antioxidant layer is 1 μm -150 μm .
4. The heating assembly of claim 1, wherein the sleeve comprises a hollow tubular body, wherein the accommodating cavity is formed in the tubular body, and wherein one end of the tubular body is provided with an opening.
5. The heating assembly of claim 4, wherein the heating portion is connected to two conductive portions, and wherein the two conductive portions penetrate through the opening.
6. The heating assembly of claim 4, wherein the sleeve comprises a pointed structure, and wherein the pointed structure is arranged at an end of the tubular body away from the opening.
7. The heating assembly of claim 5, further comprising: an insulating member at least partially arranged in the sleeve and configured to insulate the two conductive portions.
8. The heating assembly of claim 5, further comprising: a supporting base configured to support the heating portion and the sleeve, the sleeve being at least partially inserted into the supporting base; and a conductive member connected to the conductive portions in the supporting base.
9. The heating assembly of claim 8, wherein the supporting base comprises a holder configured to support the sleeve, and a seal member, and wherein the seal member is sleeved over a partial section of the sleeve and seals a gap between an inner wall of the holder and an outer wall of the sleeve.
10. The heating assembly of claim 9, wherein the seal member is a hollow structure with two ends running through, and wherein a channel for the sleeve to pass through is formed on an inner side of the hollow structure.
11. The heating assembly of claim 10, wherein the seal member comprises a sleeve body with two ends running through, the seal member being configured to be partially sleeved over the sleeve, wherein a first seal portion protrudes from an outer side wall of the sleeve body, and wherein the first seal portion is clamped and fixed with the holder.
12. The heating assembly of claim 11, wherein the supporting base comprises a shell sleeved over an outer periphery of the holder and is provided with a sleeve opening configured to cooperate with the holder, and wherein the holder comprises a bottom wall, and a gap is provided between the sleeve opening and the bottom wall.
13. The heating assembly of claim 12, wherein the shell is detachably sleeved over the holder, and wherein the shell is provided with a through hole for the heating structure to partially penetrate through.
14. The heating assembly of claim 13, wherein the seal member comprises a sleeve body with two ends running through, the seal member being configured to be partially sleeved over the sleeve, and

a second seal portion protruding from an outer side wall of the sleeve body, and wherein the second seal portion is located between the holder and the shell in an assembled state of the shell and the holder, and is configured to seal a gap formed between the holder and an end surface of the through hole.

15. The heating assembly of claim 1, wherein the sleeve comprises infrared transparent glass, transparent ceramics, or diamond.

16. The heating assembly of claim 1, wherein the heating element is spaced apart from the tube wall of the sleeve on the whole.

17. The heating assembly of claim 1, wherein the heating element is arranged without being in direct contact with the sleeve.

18. The heating assembly of claim 1, wherein the infrared radiation layer comprises an infrared layer and/or a composite infrared layer, and wherein the composite infrared layer is formed by compounding an infrared layer forming substrate with a bond configured to bond with the antioxidant layer.

19. The heating assembly of claim 1, wherein the heating substrate comprises a metal substrate, and wherein the metal substrate comprises a nickel-chromium alloy substrate or an iron-chromium-aluminum alloy substrate.

20. An aerosol generating device, comprising: the heating assembly of claim 1.
